

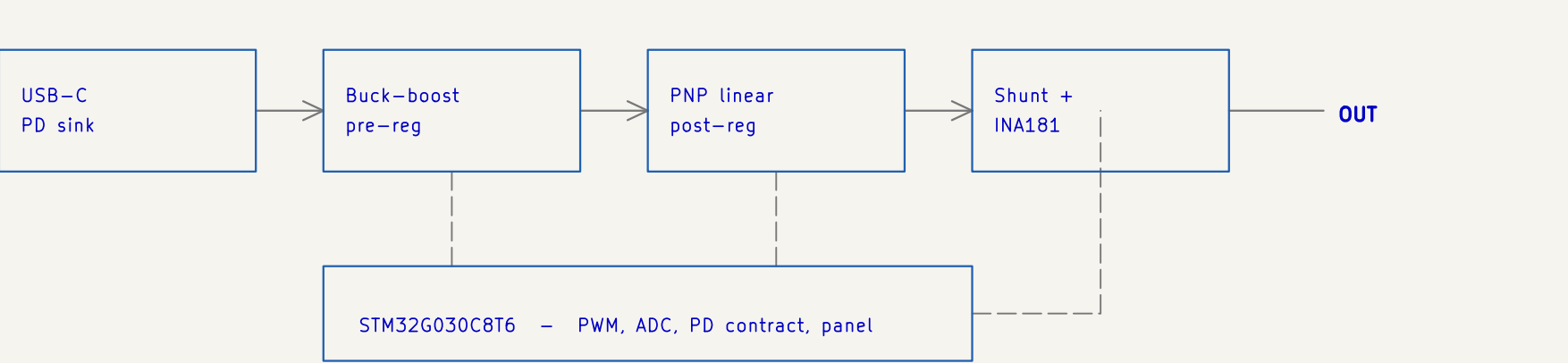
USB-C PD BENCH POWER SUPPLY

0 – 36 V / 0 – 3 A adjustable bench supply from a USB-C PD source

SPECIFICATION

Input	USB-C PD sink, 5 / 9 / 12 / 15 / 20 V contracts
Pre-regulator	4-switch synchronous buck-boost, 250 kHz
Post-regulator	PNP Darlington linear stage, tracking headroom
Output	0 – 36 V, 0 – 3 A
Set resolution	10 mV / 10 mA, PWM reference + 2-pole RC
Readback	voltage and current, 12-bit ADC
Protection	hardware over-current trip, latching
Modes	constant voltage, constant current, battery emulation
Control	STM32G030C8T6, OLED + rotary encoder + 3 keys

BLOCK DIAGRAM



REVISION HISTORY

Rev	MCU	Change
v0.3	CH32V003F4P6	first bench build; NPN Darlington follower
v0.4	CH32V003F4P6	pass device turned round to a PNP
v0.5	CH32V203C8T6	front panel added; the '003 runs out of pins
v0.6	CH32V203C8T6	hardware over-current trip
v1.1	STM32G030C8T6	redrawn as a hierarchy; on-sheet 12 V rail

DESIGNED AND DRAWN BY

Mohammadreza Safaeian

Embedded Systems & Hardware Engineer

m.re.safaeian@gmail.com

github.com/mohammadrezasafaeian

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Power Path

1 input / PD trigger
2 +12 V gate rail
3 buck-boost power stage
4 gate drivers
5 over-current trip

Control and UI

7 linear post-regulator
8 current sense
9 MCU
10 user interface
11 programming header

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Designed and drawn by Mohammadreza Safaeian – m.re.safaeian@gmail.com

Title page – specification, block diagram, sheet index

Mohammadreza Safaeian – personal project

Sheet: /

File: usbc_pd_psu.kicad_sch

Title: USB-C PD Bench PSU – 4-Switch Buck-Boost + PNP Linear P

Size: A3 Date: 2026-04-15 Rev: **v1.1**

KiCad E.D.A. 10.0.3 Id: 1/3

[illegible]

The schematic diagram illustrates a boost converter circuit. The input is a 1.2V source (C5, 47u/25V) connected to the EN pin of the MT3608 MOSFET driver (U2). The driver's SW pin is connected to an inductor (SW_BOOST), which is then connected to a diode (D2, SS34). The diode's cathode is connected to the FB pin of the driver and the positive terminal of a 12V output filter capacitor (C6, 10u/50V). The diode's anode is connected to the output of the boost converter, which is also connected to a 12V output filter capacitor (C5, 47u/25V) and a load resistor (R6, 91k 1%, R7, 4k7 1%). The output voltage is measured at TP2. The driver's GND pin is connected to ground, and its IN pin is connected to the 1.2V source.

Sheet: /Power Path/ File: power.kicad_sch	
Title: USB-C PD Bench PSU - 4-Switch Buck-Boost + PNP Linear	
Size: A3	Rev: v1.1
Date: 2026-04-15	Id: 2/3
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